# NATIONAL UNIVERSITY OF LESOTHO

**Department of Mathematics and Computer Science**
**CS4433 Software Engineering: Final Mini Project**

## PROJECT MANAGEMENT PLAN
## Moderated Learning Management System (LMS)

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COURSE: CS 4433

PROJECT: LEARNING MANAGEMENT SYSTEM

PHASE 1: DELIVERABLES

## Abstract:

This plan is building a simple LMS for a secondary school. It will be a console app written in C++. It will use My SQL to store data. This project will be build over the period of seven weeks.

This document covers Phase 1: feasibility, work breakdown, schedule, and risks.

The plan looks at whether the project is possible, compares three possible solutions, breaks the work into nine main parts, estimates the time needed, and identifies important risks.

## 1. Introduction:

## 2. Feasibility Study

Before starting the project, three possible solutions were considered.

### 2.1 Alternatives Considered

| Option | Advantages | Problems | Decision |
|---|---|---|---|
| A. HTML+CSS+C++17 + MySQL Console LMS | Works with the project requirements and can be made for the school. | Needs development time. | Selected |
| B. Open-source Web LMS | Already has many LMS features. | Needs a web server and does not meet the required C++ console project. | Rejected |
| C. Spreadsheets and Paper | Easy to start and cheap. | Poor security, no proper roles and poor record keeping. | Rejected |

### 2.2 Technical Feasibility

The technologies needed for the project are available and well known. HTML+CSS+C++ can be used to build the program. MySQL can be used to store student, course, assignment, submission and mark information. The system can use a local MySQL database so that it can work without the internet.

FEASIBLE

### 2.3 Operational Feasibility

The four user types can have different menus and permissions. The system should be simple because some teachers may not be IT experts. The menus should use simple words and the system should provide instructions. The main problem is disagreement between stakeholders, for example a

teacher wanting marks released immediately while the HoD wants to check them first. These problems should be solved before the design starts.

FEASIBLE

### 2.4 Economic Feasibility

The software tools can be obtained at no or very low cost.

| Item | Expected Cost | Reason |
|---|---|---|
| Software tools | No cash cost | Free or open-source tools can be used. |
| Hardware | No cash cost | Existing developer's laptop can be used. |
| Development | About 214 hours | This is the expected student work. |
| Maintenance | Low | The system can be maintained locally. |

The system can save time and make marking and record keeping easier.

FEASIBLE

### 2.5 Legal Feasibility

The system will store personal information about students, including minors. The system must therefore protect this information and follow the Data Protection Act, 2011 of Lesotho which deals with the protection of the citizens' critical information. Because of this only necessary information should be collected. Access should depend on the user's role. Passwords must be protected and changes to marks must be recorded.

FEASIBLE

### 2.6 Schedule Feasibility

The project has 35 working days. The expected work is about 214 hours, while about 210 hours are available. This means there is almost no extra time. The project can still be completed if progress is checked every week, less important features are removed when necessary, and some extra hours are worked during the implementation weeks.

FEASIBLE

### 2.7 Summary and Recommendation

| Area | Rating | Main Condition |
|---|---|---|
| Technical | High | Test MySQL and password security early. |
| Operational | High | Keep the interface simple. |
| Economic | High | No major cash cost. |
| Legal | Medium | Protect student information. |
| Schedule | Low to Medium | Check progress weekly and reduce scope if needed. |

## 3. Work Breakdown Structure:

The project is divided into nine main work packages. Each package produces a specific result.

| No. | Work Package | Main Result | Expected Hours | Main Work |
|---|---|---|---|---|
| 1 | Planning and Management | Project management plan | 12.3 | Planning and project control |
| 2 | Requirements | SRS, stakeholder analysis and use cases | 30.7 | Find and define requirements |
| 3 | System Design | Software design document | 40.7 | Design the system and database |
| 4 | Implementation | C++17 source code | 51.7 | Build the system |
| 5 | Testing | Tests and test results | 25.5 | Test the system |
| 6 | Configuration and QA | CI and code-quality report | 10.3 | Check code and project files |
| 7 | Deployment | Guides and release files | 10.3 | Prepare the system for use |
| 8 | Maintenance | Version 1.1 and regression tests | 12.2 | Make the required change and retest |
| 9 | Completion | Final report and viva preparation | 20.7 | Finish the project |

Total expected effort: about 214.3 hours.

### 3.1 Effort and Contingency

The most likely total work is about 209 hours, but the expected total is about 214.3 hours. Since only about 210 hours are normally available, the project has very little spare time. To help, the plan allows extra hours during Weeks 4 and 5. Less important features can also be removed if the project starts falling behind. Testing, security and the mark audit log should not be removed.

## 4. Project Schedule:

The schedule is based on the work packages. Tasks that must finish before another task starts are linked together. Tasks with no spare time are on the critical path.

### 4.1 Critical Path

The critical path is the group of tasks that cannot be delayed without delaying the whole project. It mainly goes through requirements, design, implementation, testing, deployment, maintenance and the final report. Because many tasks are on the critical path, the project has very little spare time. A delay in one important task can move the final submission date.

### 4.2 Milestones and Gates

| Milestone | Date | What Must Be Done |
|---|---|---|

| | | |
|---|---|---|
| Gate 1: Requirements | 9 Oct | Requirements and stakeholder problems are finished. |
| Gate 2: Design | 16 Oct | Design is checked and approved. |
| Gate 3: Testing | 4 Nov | Important requirements have passed testing. |
| Release v1.0 | 6 Nov | Guides, manuals and release files are ready. |
| Release v1.1 | 12 Nov | The required change is completed and tested. |
| Final Submission | 13 Nov | Final report and viva preparation are ready. |

### 4.3 Schedule Control

Progress should be checked every Friday. The developer should compare the actual work with the planned work. If an important task is more than one day late, recovery action should be taken. If it is more than two days late, less important features should be removed.

## 5. Risk Management:

A risk is something that may cause a problem in the project. Each risk is given a likelihood, impact and exposure score.

Likelihood and impact are scored from 1 to 5. Exposure is calculated as, Likelihood × Impact. A score of 15 or more is High, 8 to 14 is Medium, and 7 or less is Low.

| ID | Risk | L | I | Exposure | Level | What to Do |
|---|---|---|---|---|---|---|
| R1 | Scope keeps changing | 4 | 3 | 12 | Medium | Solve conflicts early and remove less important features. |
| R2 | Work takes longer than expected | 4 | 4 | 16 | High | Check progress weekly and work extra hours if needed. |
| R3 | Illness, power cuts or computer failure | 2 | 5 | 10 | Medium | Back up work and push files to Git regularly. |
| R4 | Developer lacks some technical skills | 3 | 3 | 9 | Medium | Make small prototypes and use documentation. |
| R5 | Security problems | 2 | 5 | 10 | Medium | Use safe database queries, protected |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | passwords and security tests. |
| R6 | System is too slow | 2 | 3 | 6 | Low | Use database indexes and test with 800 users. |
| R7 | Design, code and tests do not match | 3 | 4 | 12 | Medium | Keep the design, code and tests linked. |
| R8 | Tool or library problems | 3 | 3 | 9 | Medium | Set up tools early and keep backup options. |
| R9 | Rules are unclear | 3 | 3 | 9 | Medium | Write the rules clearly in the requirements. |
| R10 | Stakeholders are unavailable | 3 | 3 | 9 | Medium | Use the supervisor to help answer unclear questions. |

### 5.1 Main Risk

R2, which is the risk of the project taking longer than expected, is the biggest risk. It has an exposure score of 16. This is because the expected work is already slightly more than the available time. This risk should be checked every week. If the project falls behind, extra hours or removal of less important features should be used.

### 5.2 Risk Reviews

The risk list should be checked at Gate 1, Gate 2, Gate 3 and at the end of the project. New risks can be added, old risks can be closed, and risk scores can be changed.

## 6. Organisation, Monitoring and Control:

### 6.1 Roles and Responsibilities

| Role | Responsibility |
|---|---|
| Student | Acts as project manager, analyst, designer, developer, tester and QA person. Completes the work and keeps project records updated. |
| Supervisor | Checks the work at the three gates and helps answer unclear questions. |
| Stakeholders | Provide requirements and explain what the system should do. |

### 6.2 Monitoring and Contingency

Every Friday, the student should record the hours worked, how much of each task is complete, whether the project is behind schedule, and any new risks. If a critical task is more than one day late,

extra hours or changes to non-critical work should be considered. If it is more than two days late, less important features should be removed.

- Remove, lower-priority Should-have requirements.
- Never remove, Must-have requirements, user access control, password protection, mark audit records, important automated tests and the final report.

### 6.3 Change Control

After the project plan, requirements and design are approved, changes should not be made without recording the change, the reason for it, its effect on time and risk, and the final decision.